

External trigger. External triggers allow you to stabilise repetitive waveforms and make them appear static on the display by continuously displaying the same section of the input signal. Buy an oscilloscope that offers a wider range of triggering.

Generally, oscilloscopes are available with edge- and pulse-width trigger. Advance triggering options like A and B sequence, video, communication (CAN, SPI, etc) and logic triggering can help you debug faster.

Waveform capture rate. Sample rate indicates how fast an oscilloscope samples the input signal within one waveform, but the waveform capture rate refers to how quickly an oscilloscope captures a waveform. An oscilloscope with higher waveform capture rate captures fast transients better.

Number of channels. While deciding the number of channels, look at your future needs too. Increase in the number of channels adds a lot to the cost. Beware of those manufacturers who offer more channels by compromising the sample rate.

Probe specifications. Probe specifications are as important as the oscilloscope specifications because all measurements will start from the probes. The probe's bandwidth should match the scope's bandwidth. Here also the five times thumb rule will work perfectly.

The probe will be in direct contact with the circuit, so the circuit should not overload it. Otherwise, the measurement will not be correct. Resistive loading greater than 10 mega-ohms and capacitive loading less than 10 pico-farads is acceptable.

Specifications related to measurement convenience

Easy-to-use user interface, advanced automated measurements and convenient operation can really speed up your work. Below are the specifications that you need to check for measurement convenience.

Automated waveform measurements. Automated waveform measurement function helps you get numerical

readings without any hassles. Most basic digital oscilloscopes provide this function where you can choose parameters like amplitude, frequency and time period, or have all of them together on the screen. Now if you measure any waveform, the display will show the graph along with numerical values of the parameters that you have selected.

Waveform math. This function provides automatic calculations for mathematical functions like integration, differentiation, FFT and logarithm.

User interface. The oscilloscope should be easy to use with an intuitive menu so that even a beginner can immediately get started with his work. With dedicated knobs for frequently used functions, reading the waveforms becomes handy.

Convenience of waveform navigation and analysis is highly important as, at the end of the day, you would want to navigate the waveform that you have captured, and would also want to analyse it to extract useful information. Look for options like zoom and pan, interesting mark points and search options. Such features reduce the time to debug applications to a great extent.

In-built automated test. In-built automated tests like pass/fail test are sometimes very useful to quickly identify a problem. Such tests monitor your system continuously and inform about an error through a message on the display or some sound. Some oscilloscopes also log the history of these test results, which can be very handy to analyse the errors that occurred in the system.

Connectivity. Connectivity of an oscilloscope is highly important as you will always need to analyse and share the measurements. It can also help a lot while documenting your work. A lot of interfaces like GPIB, Ethernet, RS-232 and USB are available in oscilloscopes. Some advanced oscilloscopes also let you edit your documents and print them, or send via the Internet.

Specifications related to physical construction

Buy an oscilloscope that suits your work environment. For harsh environ-

An ISO 9001:2008 Certified

Crisp System



 Arduino Kit	 Fpga Trainer Kit
 DSP Trainer	 Fingerprint Module
 Force Sensor	 Color Sensor
 GSM Modem	 Voice Recognition Kit
 MP3 Module	 Versa Pad
 RF Transceiver	 Wi-Fi Transceiver
 Accelerometer Sensor Module	 Thermoelectric Power Generator
 Fire Sensor	 Peltier Device
 Humidity Sensor Module	 Pulse Rate Sensor Module
 Voice Recording and Playback Module	 Power line Communication Module
 RFID Reader Module	 Gas Sensor Module

All the above Modules are Ready to Interface with any Micro controller

Crisp Towers, 120, Nehru Street, Ramnagar
 Coimbatore - 641 009
 Call : 7373 63 0000 / 98422 18968
 Email : sales@crispprojects.com
 Web : www.crispprojects.com

Project Guidance also available